Highlights

FedRAS: Network-Conditioned Rank Allocation for Federated LoRA

- LoRA rank is treated as a deadline-aware, network-conditioned schedulable resource
- Client admission and rank selection are jointly scheduled under a byte budget
- An exact grid dynamic program solves the assignment at sub-second per-round cost
- Frozen-factor rank masking aggregates heterogeneous ranks exactly
- 22% more timely updates at 23% fewer bytes with statistically equal accuracy